$$J = \begin{bmatrix} 0 & L_t\cos(\theta_t) + L_s\sin(\theta_t-\theta_s) & -L_s\sin(\theta+\epsilon-\theta_s) \\ -L_h\sin(\theta_h) - \cos(\theta_h)\cdot\left(L_t\cos(\theta_t) + L_s\sin(\theta_t-\theta_s)\right) & L_t\sin(\theta_t)\sin(\theta_h) - L_s\sin(\theta_h)\cos(\theta_t-\theta_s) & L_s\sin(\theta_h)\cos(\theta_t-\theta_s \\ L_h\cos(\theta_h) - \sin(\theta_h)\left(L_t\cos(\theta_t) + L_s\sin(\theta_t-\theta_s)\right) & L_t\sin(\theta_t)\cos(\theta_h) + L_s\cos(\theta_h)\cos(\theta_h-\theta_t) & -L_s\cos(\theta_h) \\ & & \cos(\theta_t-\theta_s) \end{bmatrix}$$